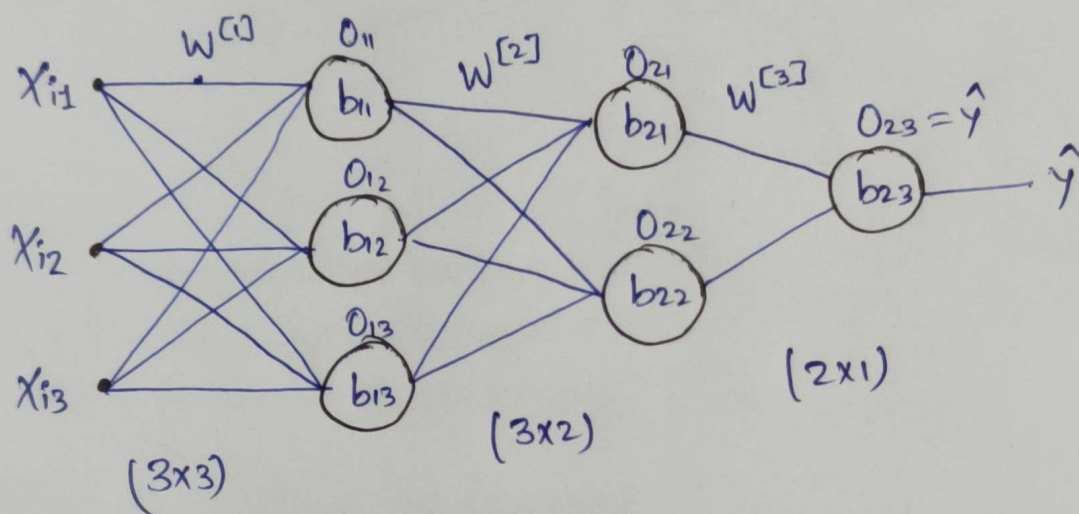


→ Memorization

It is the process of learning and retaining information in memory so that it can be recalled later.

MLP Memorization



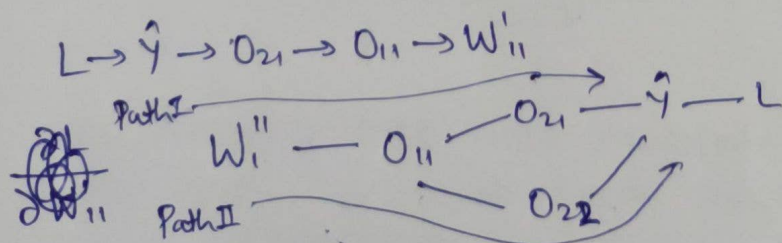
$$L \rightarrow \hat{y} \rightarrow W_{11}^3$$

$$\frac{\partial L}{\partial W_{11}^3} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial W_{11}^3}$$

$$L \rightarrow \hat{y} \rightarrow o_{21} \rightarrow W_{11}^2$$

$$\frac{\partial L}{\partial W_{11}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial o_{21}} \times \frac{\partial o_{21}}{\partial W_{11}^2}$$

Here we store the derivative
the 2 memoization is used
for faster the in finding
the derivative.



$$\frac{\partial L}{\partial W'_{11}} = \frac{\partial L}{\partial \hat{y}} \left[\frac{\partial \hat{y}}{\partial o_{21}} \times \frac{\partial o_{21}}{\partial o_{11}} \times \frac{\partial o_{11}}{\partial W'_{11}} + \frac{\partial \hat{y}}{\partial o_{22}} \times \frac{\partial o_{22}}{\partial o_{11}} \times \frac{\partial o_{11}}{\partial W'_{11}} \right]$$

Back prop → chain diff(rule) + Memorization.